



SALAS O'BRIEN
3200 WINDY HILL ROAD SE
SUITE 200E
ATLANTA, GEORGIA 30339
PHONE 404-881-5300

Project: Wehunt Preserve; Davidson Homes
Location: Braselton, GA
Description: Ash - B, Slab, Bed4, Loft

Date: August 8, 2024
By: JTH / TJH
House Faces: Worst NorthEast

These Loads are dependent on the information/assumptions listed below. **Any deviation from these assumptions during construction could have a significant impact on the cooling and heating loads for this house and render the recommended equipment sizing invalid.** If any of these assumptions are incorrect, please let us know immediately so we can re-calculate them. This information will be used in a residential heating and cooling load program based on calculations recommended by ASHRAE, Handbook of Fundamentals. Equipment sizing indicated in this report is valid only upon verification that actual construction is in full compliance with the descriptions below and independent testing of HVAC systems has been performed; such verification and testing shall be performed by Vallus.

BUILDING DATA	
Typical Ceiling Height	9.0 ft
Floor Area	2,625 ft ²
Volume	23,160 ft ³
Percentage Glass	11 %
Number of Bedrooms	4

DESIGN CONDITIONS		
	Summer	Winter
Outdoor	95 °Fdb	18 °Fdb
Indoor	75 °Fdb	72 °Fdb
ΔT	20 °Fdb	54 °Fdb
Range	17.3 °Fdb	

ENERGY CALCULATIONS	
Right Choice HVAC kWh	9689
SEER	14
ACH50	5.0
Mechanical Ventilation	NO
House Outside Air CFM	N/A

CONSTRUCTION DESCRIPTIONS						
Walls / Partitions	#	U-Value	Description/Notes			
	W1	0.077	Above Grade	2x4	R-13 batt	
	W2	0.067	Basement	Concrete + 2x4	R-13 batt	
	W3	0.053	Attic	2x6	R-19 batt	
Doors	#	U-Value	Description/Notes			
	D1	0.13	Exterior Door		R-7.7	
	D2	0.083	Garage Door		R-12	
Roofs	#	U-Value	Description/Notes			
	R1	0.026	Flat Ceiling		R-38 blown	
	R2	0.026	Vaulted		R-38 batt	
Floors	#	U-Value	Description/Notes			
	F1	0.026	Framed		R-38 batt	
	F2	0.1	Slab		None	
Glass	#	Glass Type	Windows	U-Value	Internal Shading	Notes
	G1	Double Insulated	All types	0.35	Blinds/Draperies	SHGC 0.22
	G2					
	G3					

LOAD SUMMARY					
		Unit 1 - Whole House			
		Cooling		Heating	
Sensible Load		27666 Btu/hr		32237 Btu/hr	
Minimum Nominal Capacity		3.07 Tons		9.45 kW	
Load Density		854 SF/Ton		3.6 W/SF	
Floor Area Served		2625.0 SF		2625.0 SF	
Mechanical Ventilation		N/A CFM		N/A CFM	
System Size		3.0 Tons		15.0 kW	
Leaving Air Temperature		60.1 deg F		103.9 deg F	
Installed Capacity		875 SF/Ton		5.7 W/SF	

I understand the information provided herein, and warrant, to the best of my ability and knowledge, and to the full extent of my control as the Contractor, that this house will be built in accordance with the information given above.

Signed _____ Title _____ Date _____

Project:	Wehunt Preserve; Davidson Homes
Location:	Braselton, GA
Description:	Ash - B, Slab, Bed4, Loft

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[illegible]

Unit 1

Size

3.00

Tons

Airflow

1200

CFM

Heat kW

15

kW

First Floor Airflows

Room	CFM Cool	CFM Heat	CFM Avg	Duct Size
Foyer	23	39	31	
Dining	66	125	96	
Kitchen	159	92	126	
Breakfast	69	98	84	
Family	167	130	149	
Owner's Suite	108	127	118	
Owner's Bath	26	48	37	
Owner's WIC	24	37	31	
Laundry	45	16	31	
Mud Rm	12	16	14	
Powder	4	8	6	
-	-	-	-	
-	-	-	-	
-	-	-	-	

Unit 1 (Continued)				
Second Floor Airflows				
Room	CFM Cool	CFM Heat	CFM Avg	Duct Size
Stairs	15	25	20	
Hallway & Alcove	38	46	42	
Bed 2	72	89	80	
Bath 2	17	20	18	
Bed 3	90	80	85	
Bath 3	34	28	31	
Bed 4	81	69	75	
Bed 4 WIC	9	12	11	
Loft	140	94	117	
-	-	-	-	
-	-	-	-	
-	-	-	-	
-	-	-	-	
-	-	-	-	

Cooling Size	2	DX Cooling kWh	2165
Heating Size	2	RC Heating kWh	3687
Best Case	2	Total kWh	5851

[illegible]

Project:

Location:

Description:

Whewunt Preserve; Davidson Homes

Braselton, GA

Ash - B, Slab, Bed4, Loft

Date: August 8, 2024

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Level(s): Second Floor

Square Feet

1145

Volume

9840

SF/Ton

763

SEER

14

Typical Ceiling Height	8	Cooling Size	1.5	DX Cooling kWh	1524
Kneewall Height	1	Heating Size	1.5	RC Heating kWh	2314
		Best Case	1.5	Total kWh	3838

COOLING LOAD CALCULATION										Ceiling Height		18		8		8		8		8		8		8		8		8		8		8			
Components					CLTD (°F)	U-Value (Btu/hr-sf-F)	Cooling Load Factor	Entire House		Stairs		Hallway & Alcove		Bed 2		Bath 2		Bed 3		Bath 3		Bed 4		Bed 4 WIC		Loft									
Type	Description			Qty				Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	
Glass	G1	NorthWest	Right	(E)	18	Btu/hr-sf	30	552													15	276			15	276									
	G1	North		(SE)	6	Btu/hr-sf	0	0																											
	G1	NorthEast	Front	(S)	8	Btu/hr-sf	42	329				12	94	30	235																				
	G1	East		(SW)	13	Btu/hr-sf	0	0																											
	G1	SouthEast	Left	(W)	14	Btu/hr-sf	15	214										15	214																
	G1	South		(NW)	16	Btu/hr-sf	0	0																											
	G1	SouthWest	Back	(N)	26	Btu/hr-sf	0	0																											
	G1	West		(NE)	29	Btu/hr-sf	0	0																											
	G1	Sky Light			46	Btu/hr-sf	0	0																											
	G1	Shaded			6	Btu/hr-sf	0	0																											
Door	D1	Door		21	0.13	2.73	Btu/hr-sf	0	0																										
	D2	Door		55	0.08	4.57	Btu/hr-sf	18	81			17.8	81																						
Above Grade Walls	W1	NorthWest		19	0.08	1.46	Btu/hr-sf	270	395	-	-	34	50	18	26	-	-	-	-	-	-	98	143	40	58	80	117	-	-	-	-	-	-		
	W1	North		13	0.08	1.00	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	NorthEast		19	0.08	1.46	Btu/hr-sf	218	319	-	-	42	61	84	123	-	-	-	-	-	-	-	-	-	-	92	134	-	-	-	-	-			
	W1	East		23	0.08	1.77	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	SouthEast		21	0.08	1.62	Btu/hr-sf	287	464	-	-	-	-	124	200	52	84	112	181	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	South		16	0.08	1.23	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	SouthWest		21	0.08	1.62	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	West		23	0.08	1.77	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	NorthWest		19	0.08	1.46	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	North		13	0.08	1.00	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
Basement Walls	W1	NorthEast		19	0.08	1.46	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	East		23	0.08	1.77	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	SouthEast		21	0.08	1.62	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	South		16	0.08	1.23	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	SouthWest		21	0.08	1.62	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	West		23	0.08	1.77	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W2	Grade		0	0.07	0.00	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	Partition		10	0.08	0.77	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	Garage		30	0.08	2.31	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W3	Attic		55	0.05	2.92	Btu/hr-sf	558	1626	-	-	12	35	-	-	-	182	531	151	439	121	352	-	-	92	268	-	-	-	-	-	-	-		
Other	W3	Kneewall		55	0.05	2.92	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	R1	Ceiling		55	0.03	1.43	Btu/hr-sf	1132	1619	68	97	124	177	172	246	72	103	166	237	69	99	172	246	35	50	254	363	-	-	-	-	-	-		
	R2	Vaulted		55	0.03	1.43	Btu/hr-sf	18	26	-	-	-	-	-	-	-	13	26	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	F1	Crawl/Uncon.		11	0.03	0.29	Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
Floor	F1	Garage		30	0.03	0.78	Btu/hr-sf	61	48	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	61	48	-	-	-	-	-	-	-		
	F2	Slab		0	0.10	0.00	Btu/hr-sf	67	0	67	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
Internal	People					255	Btu/hr	4	1020	-	-	-	1	255	-	-	1	255	-	-	1	255	-	-	1	255	-	-	-	-	-	-	-		
	Lights (W)					3.413	Btu/hr-W	574	1959	34	116	62	212	86	294	36	123	90	307	35	119	86	294	18	61	127	433	-	-	-	-	-	-		
	Appliances (W)					3.413	Btu/hr-W	250	853	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	250	853	-	-	-	-	-	-	-		
	Infiltration					0.25	ACH	0.09	Btu/hr-cf	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
Vent	Volume					CF		9840		1224		992		1376		576		1432		552		1376		280		2032		0		0		0			
	Area					SF		1145		68		124		172		72		179		69		172		35		254		0		0		0			
Subtotal (Btu/hr)									10391		323		801		1503		361		1881		707		1689		195		2931		0		0		0		
Safety Factor									1039																										
Sensible Cooling Load (Btu/hr)									11430																										
Nominal Tons Required (75% sens. cap.)									1.27																										
System Selection 1.5 Tons									600		19		46		87		21		109		41		98		11		169		-		-		-		
Space CFM									600																										
HEATING LOAD CALCULATION																																			
Components					CLTD (°F)	U-Value (Btu/hr-sf-F)	Heating Load Factor	Entire House		Stairs		Hallway & Alcove		Bed 2		Bath 2		Bed 3		Bath 3		Bed 4		Bed 4 WIC		Loft		0		0		0		0	
Type	Description				Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	
Glass	G1			54	0.35	18.9	Btu/hr-sf	87	1644			12	227	30	567	0	-	15	284	0	-	15	284	0	-	15	284	0	-	0	-	0	-	0	-
	G2			54	0.00	0.0	Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-
Door	G3			54	0.00	0.0	Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-
	D1			54	0.13	7.0	Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-
Walls	D2			54	0.08	4.5	Btu/hr-sf	18	80	0	-	17.8	80	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-
	W1			54	0.08	4.2	Btu/hr-sf	775	3223	0	-	76.5	318	226	939	51.8	215	112	465	0	-	97.5	405	39.8	165	172	716	0	-	0	-	0	-	0	-
	W2			54	0.07	3.6	Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-
	W3			54	0.05	2.9	Btu/hr-sf	558	1597	0	-	12.2	35	0	-	0	-	182																	